

Fast Macaulay Null Space

Nithin Govindarajan

with Raphaël Widdershoven and Lieven De Lathauwer

September 11th, 2025

Workshop on Low-Rank Models and Applications (LRMA 25)

Outline

Systems of multivariate polynomial equations

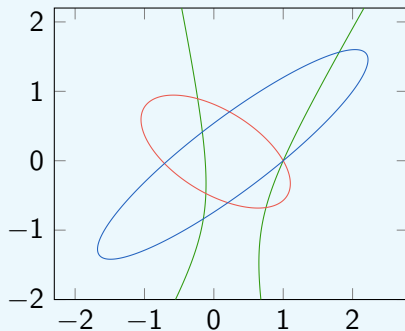
Macaulay-based solution method

Fast Macaulay null space

Resulting timing and complexity

Systems of multivariate polynomial equations

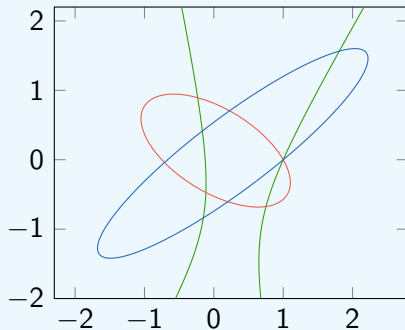
$$\begin{cases} -3 - x - 2y + 4x^2 + 6xy + 7y^2 = 0 \\ -2 - x + y + 3x^2 - 7xy + 5y^2 = 0 \\ 1 + 7x + y - 8x^2 + 3xy + y^2 = 0 \end{cases}$$



Systems of multivariate polynomial equations

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- $N = 2$ unknowns
- $S = 3$ equations $\rightarrow$ overdetermined
- Degree $D = 2$



Examples

Special cases:

- Linear systems: $\mathbf{Ax} = \mathbf{b}$
- Eigenvalue problems: $\mathbf{Ax} = \lambda \mathbf{x}$
- Most tensor problems, e.g. CPD: $\mathcal{T} = \sum_{r=1}^R \mathbf{a}_r^{(1)} \otimes \cdots \otimes \mathbf{a}_r^{(N)}$

Examples

Special cases:

- Linear systems: $\mathbf{Ax} = \mathbf{b}$
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Application fields:

- Kinematics (using trig identities and $c_n = \cos(\theta_n)$, $s_n = \sin(\theta_n)$):

$$\begin{cases} x = L_1 \cos(\theta_1) + L_2 \cos(\theta_1 + \theta_2) \\ y = L_1 \sin(\theta_1) + L_2 \sin(\theta_1 + \theta_2) \end{cases}$$

- Camera geometry
- Cryptography

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Solution idea

$$\mathbf{\Sigma} = \begin{cases} p_1(x, y) \\ p_2(x, y) \\ p_3(x, y) \end{cases} = \begin{cases} -3 - x - 2y + 4x^2 + 6xy + 7y^2 \\ -2 - x + y + 3x^2 - 7xy + 5y^2 \\ 1 + 7x + y - 8x^2 + 3xy + y^2 \end{cases} = \mathbf{0}$$

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$$\left[\begin{array}{ccc|ccc} -3 & -1 & -2 & 4 & 6 & 7 \\ -2 & -1 & 1 & 3 & -7 & 5 \\ 1 & 7 & 1 & -8 & 3 & 1 \end{array} \right] \begin{bmatrix} 1 \\ x \\ y \\ x^2 \\ xy \\ y^2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\mathbf{C}_2 \mathbf{v}_2 = \mathbf{0}$$

Solution idea

$$\Sigma = \begin{cases} p_1(x, y) \\ p_2(x, y) \\ p_3(x, y) \end{cases} = \begin{cases} -3 - x - 2y + 4x^2 + 6xy + 7y^2 \\ -2 - x + y + 3x^2 - 7xy + 5y^2 \\ 1 + 7x + y - 8x^2 + 3xy + y^2 \end{cases} = \mathbf{0}$$

$$\left[\begin{array}{ccc|ccc} -3 & -1 & -2 & 4 & 6 & 7 \\ -2 & -1 & 1 & 3 & -7 & 5 \\ 1 & 7 & 1 & -8 & 3 & 1 \end{array} \right] \begin{bmatrix} 1 \\ x \\ y \\ x^2 \\ xy \\ y^2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$
$$\mathbf{C}_2 \mathbf{v}_2 = \mathbf{0}$$

Problems:

- Additional “spurious” vectors may appear

Solution idea

$$\Sigma = \begin{cases} p_1(x, y) \\ p_2(x, y) \\ p_3(x, y) \end{cases} = \begin{cases} -3 - x - 2y + 4x^2 + 6xy + 7y^2 \\ -2 - x + y + 3x^2 - 7xy + 5y^2 \\ 1 + 7x + y - 8x^2 + 3xy + y^2 \end{cases} = \mathbf{0}$$

$$\begin{bmatrix} -3 & -1 & -2 & 4 & 6 & 7 \\ -2 & -1 & 1 & 3 & -7 & 5 \\ 1 & 7 & 1 & -8 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ x \\ y \\ x^2 \\ xy \\ y^2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$
$$\mathbf{C}_2 \mathbf{v}_2 = \mathbf{0}$$

Problems:

- Additional “spurious” vectors may appear
- Multiple roots $\rightarrow$ linear combinations of Vandermonde vectors

Macaulay matrix

$$\begin{bmatrix} p_1(x, y) \\ p_2(x, y) \\ p_3(x, y) \\ \hline x p_1(x, y) \\ x p_2(x, y) \\ x p_3(x, y) \\ \hline y p_1(x, y) \\ y p_2(x, y) \\ y p_3(x, y) \end{bmatrix} = \begin{bmatrix} -3 & -1 & -2 & 4 & 6 & 7 & 0 & 0 & 0 & 0 \\ -2 & -1 & 1 & 3 & -7 & 5 & 0 & 0 & 0 & 0 \\ 1 & 7 & 1 & -8 & 3 & 1 & 0 & 0 & 0 & 0 \\ \hline 0 & -3 & 0 & -1 & -2 & 0 & 4 & 6 & 7 & 0 \\ 0 & -2 & 0 & -1 & 1 & 0 & 3 & -7 & 5 & 0 \\ 0 & 1 & 0 & 7 & 1 & 0 & -8 & 3 & 1 & 0 \\ \hline 0 & 0 & -3 & 0 & -1 & -2 & 0 & 4 & 6 & 7 \\ 0 & 0 & -2 & 0 & -1 & 1 & 0 & 3 & -7 & 5 \\ 0 & 0 & 1 & 0 & 7 & 1 & 0 & -8 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ \hline x \\ y \\ \hline x^2 \\ xy \\ y^2 \\ \hline x^3 \\ x^2y \\ xy^2 \\ y^3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \hline 0 \\ 0 \\ 0 \\ \hline 0 \\ 0 \\ 0 \end{bmatrix}$$

$\mathbf{M}_3 \mathbf{v}_3 = \mathbf{0}$

Macaulay matrix

$$\begin{bmatrix} p_1(x, y) \\ p_2(x, y) \\ p_3(x, y) \\ \hline x p_1(x, y) \\ x p_2(x, y) \\ x p_3(x, y) \\ \hline y p_1(x, y) \\ y p_2(x, y) \\ y p_3(x, y) \end{bmatrix} = \begin{bmatrix} -3 & -1 & -2 & 4 & 6 & 7 & 0 & 0 & 0 & 0 \\ -2 & -1 & 1 & 3 & -7 & 5 & 0 & 0 & 0 & 0 \\ 1 & 7 & 1 & -8 & 3 & 1 & 0 & 0 & 0 & 0 \\ \hline 0 & -3 & 0 & -1 & -2 & 0 & 4 & 6 & 7 & 0 \\ 0 & -2 & 0 & -1 & 1 & 0 & 3 & -7 & 5 & 0 \\ 0 & 1 & 0 & 7 & 1 & 0 & -8 & 3 & 1 & 0 \\ \hline 0 & 0 & -3 & 0 & -1 & -2 & 0 & 4 & 6 & 7 \\ 0 & 0 & -2 & 0 & -1 & 1 & 0 & 3 & -7 & 5 \\ 0 & 0 & 1 & 0 & 7 & 1 & 0 & -8 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ \hline x \\ y \\ \hline x^2 \\ xy \\ y^2 \\ \hline x^3 \\ x^2y \\ xy^2 \\ y^3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \hline 0 \\ 0 \\ 0 \\ \hline 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$\mathbf{M}_3 \mathbf{v}_3 = \mathbf{0}$

- Starting at degree E , the Macaulay matrix has a fixed null space dimension

Recovery of roots from null space

- Given: a basis $\mathbf{N}_E$ for the null space of Macaulay matrix $\mathbf{M}_E$

- Known: $\mathbf{N}_E = \mathbf{V}_E \mathbf{G}$ for some square $\mathbf{G}$ and $\mathbf{V}_E =$

$$\begin{bmatrix} 1 & \dots & 1 \\ \hline x^{(1)} & \dots & x^{(L)} \\ y^{(1)} & \dots & y^{(L)} \\ \hline (x^{(1)})^2 & \dots & (x^{(L)})^2 \\ x^{(1)}y^{(1)} & \dots & x^{(L)}y^{(L)} \\ (y^{(1)})^2 & \dots & (y^{(L)})^2 \\ \hline \vdots & \vdots & \vdots \end{bmatrix}$$

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- Solution: compute rank- L CPD:

$$\text{cat}_3(\mathbf{S}_1 \mathbf{N}_E, \mathbf{S}_x \mathbf{N}_E, \mathbf{S}_y \mathbf{N}_E) = \llbracket \mathbf{V}_{E-1}, \mathbf{G}, \mathbf{X} \rrbracket$$

with selection matrices $\mathbf{S}_1, \mathbf{S}_x, \mathbf{S}_y$ and solution matrix $\mathbf{X} = \begin{bmatrix} 1 & \dots & 1 \\ x^{(1)} & \dots & x^{(L)} \\ y^{(1)} & \dots & y^{(L)} \end{bmatrix}$.

Macaulay-based solution of polynomial systems

Algorithm 1 Macaulay-based solution of polynomial systems.

Require: A system Σ_D which has a finite number of roots.

Ensure: Matrix $\mathbf{X}$ which contains the roots of Σ_D .

- 1: Create the Macaulay matrix $\mathbf{M}_E$ associated with Σ_D of sufficient degree E .
 - 2: Compute $\mathbf{N}_E = \text{null}(\mathbf{M}_E)$.
 - 3: Form $\mathcal{C}$ by determining and stacking $\mathbf{S}_1 \mathbf{N}_E, \mathbf{S}_{x_1} \mathbf{N}_E, \dots, \mathbf{S}_{x_N} \mathbf{N}_E$.
 - 4: Compute the CPD of $\mathcal{C} = [\mathbf{V}_{E-1}, \mathbf{G}, \mathbf{X}]$.
-

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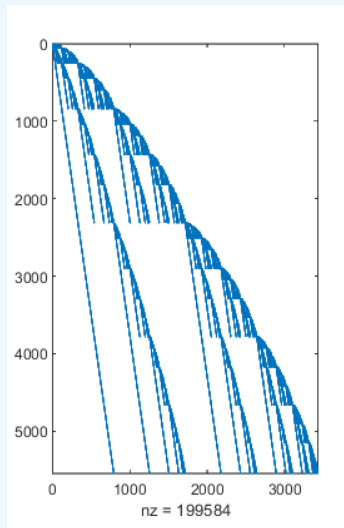
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Macaulay matrices are very large, sparse and structured

Degree $D = 2$
 $N = 7$ variables
 $S = 7$ equations
 $D^N = 128$ roots



Example system

$$\begin{bmatrix} p_1 \\ p_2 \\ x p_1 \\ x p_2 \\ y p_1 \\ y p_2 \\ x^2 p_1 \\ x^2 p_2 \\ xy p_1 \\ xy p_2 \\ y^2 p_1 \\ y^2 p_2 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 & 1 & 5 & 6 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 2 & 1 & 7 & 8 & 1 & 9 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 2 & 3 & 0 & 1 & 5 & 6 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 1 & 7 & 0 & 8 & 1 & 9 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 2 & 3 & 0 & 1 & 5 & 6 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 1 & 7 & 0 & 8 & 1 & 9 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 2 & 3 & 0 & 0 & 1 & 5 & 6 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 1 & 7 & 0 & 0 & 8 & 1 & 9 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 2 & 3 & 0 & 0 & 1 & 5 & 6 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 & 7 & 0 & 0 & 8 & 1 & 9 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 2 & 3 & 0 & 0 & 1 & 5 & 6 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 & 7 & 0 & 0 & 8 & 1 & 9 \end{bmatrix} \begin{bmatrix} 1 \\ x \\ y \\ x^2 \\ xy \\ y^2 \\ x^3 \\ x^2y \\ xy^2 \\ y^3 \\ x^4 \\ x^3y \\ x^2y^2 \\ xy^3 \\ y^4 \end{bmatrix}$$

Null space through intersections

$$\text{null}(\mathbf{M}_2) = \text{null} \left(\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 5 & 6 \\ 2 & 1 & 7 & 8 & 1 & 9 \end{array} \right] \right)$$

Null space through intersections

$$\text{null}(\mathbf{M}_3) = \text{null} \left(\left[\begin{array}{ccc|ccc|cccc} 1 & 2 & 3 & 1 & 5 & 6 & 0 & 0 & 0 & 0 \\ 2 & 1 & 7 & 8 & 1 & 9 & 0 & 0 & 0 & 0 \\ \hline 0 & 1 & 0 & 2 & 3 & 0 & 1 & 5 & 6 & 0 \\ 0 & 2 & 0 & 1 & 7 & 0 & 8 & 1 & 9 & 0 \\ 0 & 0 & 1 & 0 & 2 & 3 & 0 & 1 & 5 & 6 \\ 0 & 0 & 2 & 0 & 1 & 7 & 0 & 8 & 1 & 9 \end{array} \right] \right)$$

Null space through intersections

$$\begin{aligned}
 \text{null}(\mathbf{M}_3) &= \text{null} \left(\left[\begin{array}{ccc|ccc|cccc} 1 & 2 & 3 & 1 & 5 & 6 & 0 & 0 & 0 & 0 \\ 2 & 1 & 7 & 8 & 1 & 9 & 0 & 0 & 0 & 0 \\ \hline 0 & 1 & 0 & 2 & 3 & 0 & 1 & 5 & 6 & 0 \\ 0 & 2 & 0 & 1 & 7 & 0 & 8 & 1 & 9 & 0 \\ 0 & 0 & 1 & 0 & 2 & 3 & 0 & 1 & 5 & 6 \\ 0 & 0 & 2 & 0 & 1 & 7 & 0 & 8 & 1 & 9 \end{array} \right] \right) \\
 &= \text{null} \left(\left[\begin{array}{ccc|ccc|cccc} 1 & 2 & 3 & 1 & 5 & 6 & 0 & 0 & 0 & 0 \\ 2 & 1 & 7 & 8 & 1 & 9 & 0 & 0 & 0 & 0 \end{array} \right] \right) \\
 &\cap \text{null} \left(\left[\begin{array}{ccc|ccc|cccc} 0 & 1 & 0 & 2 & 3 & 0 & 1 & 5 & 6 & 0 \\ 0 & 2 & 0 & 1 & 7 & 0 & 8 & 1 & 9 & 0 \end{array} \right] \right) \\
 &\cap \text{null} \left(\left[\begin{array}{ccc|ccc|cccc} 0 & 0 & 1 & 0 & 2 & 3 & 0 & 1 & 5 & 6 \\ 0 & 0 & 2 & 0 & 1 & 7 & 0 & 8 & 1 & 9 \end{array} \right] \right).
 \end{aligned}$$

Null space through intersections

$$\begin{aligned}
 & \left[\begin{array}{c|cc|ccc|cccc} 1 & x & y & x^2 & xy & y^2 & x^3 & x^2y & xy^2 & y^3 \end{array} \right] \\
 \text{null}(\mathbf{M}_3) = & \text{null} \left(\left[\begin{array}{ccc|ccc|cccc} 1 & 2 & 3 & 1 & 5 & 6 & 0 & 0 & 0 & 0 \\ 2 & 1 & 7 & 8 & 1 & 9 & 0 & 0 & 0 & 0 \\ \hline 0 & 1 & 0 & 2 & 3 & 0 & 1 & 5 & 6 & 0 \\ 0 & 2 & 0 & 1 & 7 & 0 & 8 & 1 & 9 & 0 \\ 0 & 0 & 1 & 0 & 2 & 3 & 0 & 1 & 5 & 6 \\ 0 & 0 & 2 & 0 & 1 & 7 & 0 & 8 & 1 & 9 \end{array} \right] \right) \\
 = & \text{null} \left(\left[\begin{array}{ccc|ccc|cccc} 1 & 2 & 3 & 1 & 5 & 6 & 0 & 0 & 0 & 0 \\ 2 & 1 & 7 & 8 & 1 & 9 & 0 & 0 & 0 & 0 \end{array} \right] \right) \\
 \cap & \text{null} \left(\left[\begin{array}{ccc|ccc|cccc} 0 & 1 & 0 & 2 & 3 & 0 & 1 & 5 & 6 & 0 \\ 0 & 2 & 0 & 1 & 7 & 0 & 8 & 1 & 9 & 0 \end{array} \right] \right) \\
 \cap & \text{null} \left(\left[\begin{array}{ccc|ccc|cccc} 0 & 0 & 1 & 0 & 2 & 3 & 0 & 1 & 5 & 6 \\ 0 & 0 & 2 & 0 & 1 & 7 & 0 & 8 & 1 & 9 \end{array} \right] \right).
 \end{aligned}$$

“Shifting” null spaces with 1

$$\text{null} \left(\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 5 & 6 \\ 2 & 1 & 7 & 8 & 1 & 9 \end{array} \right] \right) = \text{col} \left(\left[\begin{array}{cccc} -2 & 1 & -4 & -2 \\ 0 & -3 & -1 & -1 \\ 3 & 0 & 0 & -3 \\ \hline -2 & 1 & 1 & 2 \\ -1 & 2 & 1 & 1 \\ 0 & -1 & 0 & 1 \end{array} \right] \right)$$

$$\Rightarrow \text{null} \left(\left[\begin{array}{ccc|ccc|ccc} 1 & 2 & 3 & 1 & 5 & 6 & 0 & 0 & 0 \\ 2 & 1 & 7 & 8 & 1 & 9 & 0 & 0 & 0 \end{array} \right] \right) = \text{col} \left(\left[\begin{array}{cccc|ccc} -2 & 1 & -4 & -2 & 0 & 0 & 0 \\ 0 & -3 & -1 & -1 & 0 & 0 & 0 \\ 3 & 0 & 0 & -3 & 0 & 0 & 0 \\ \hline -2 & 1 & 1 & 2 & 0 & 0 & 0 \\ -1 & 2 & 1 & 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{array} \right] \right)$$

"Shifting" null spaces with x

$$\text{null} \left(\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 5 & 6 \\ 2 & 1 & 7 & 8 & 1 & 9 \end{array} \right] \right) = \text{col} \left(\left[\begin{array}{cccc} -2 & 1 & -4 & -2 \\ 0 & -3 & -1 & -1 \\ 3 & 0 & 0 & -3 \\ \hline -2 & 1 & 1 & 2 \\ -1 & 2 & 1 & 1 \\ 0 & -1 & 0 & 1 \end{array} \right] \right)$$

$$\Rightarrow \text{null} \left(\left[\begin{array}{ccc|ccc|ccc} 0 & 1 & 0 & 2 & 3 & 0 & 1 & 5 & 6 \\ 0 & 2 & 0 & 1 & 7 & 0 & 8 & 1 & 9 \end{array} \right] \right) = \text{col} \left(\left[\begin{array}{cccc|ccc} 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ \hline -2 & 1 & -4 & -2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ \hline 0 & -3 & -1 & -1 & 0 & 0 & 0 \\ 3 & 0 & 0 & -3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ \hline -2 & 1 & 1 & 2 & 0 & 0 & 0 \\ -1 & 2 & 1 & 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 & 0 & 0 & 0 \end{array} \right] \right)$$

Intersecting null spaces

$$\begin{aligned}
 & \text{col} \left(\left[\begin{array}{cccc|ccc} 0 & -3 & -1 & -1 & 0 & 0 & 0 \\ -2 & 1 & 1 & 2 & 0 & 0 & 0 \\ -1 & 2 & 1 & 1 & 0 & 0 & 0 \\ \hline -2 & 1 & -4 & -2 & 0 & 0 & 0 \\ 3 & 0 & 0 & -3 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{array} \right] \right) \cap \text{col} \left(\left[\begin{array}{cccc|ccc} -2 & 1 & -4 & -2 & 0 & 0 & 0 \\ 0 & -3 & -1 & -1 & 0 & 0 & 0 \\ 3 & 0 & 0 & -3 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ \hline -2 & 1 & 1 & 2 & 0 & 0 & 0 \\ -1 & 2 & 1 & 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 & 0 & 0 & 0 \end{array} \right] \right) \\
 &= \text{col} \left(\left[\begin{array}{cc} \mathbf{C}_1 & \mathbf{0} \\ \mathbf{D}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{I} \end{array} \right] \right) \cap \text{col} \left(\left[\begin{array}{cc} \mathbf{C}_2 & \mathbf{0} \\ \mathbf{0} & \mathbf{I} \\ \mathbf{D}_2 & \mathbf{0} \end{array} \right] \right) \\
 &= \left[\begin{array}{cc} \mathbf{C}_1 & \mathbf{0} \\ \mathbf{D}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{D}_2 \end{array} \right] \text{null} \left(\left[\mathbf{C}_1 \quad -\mathbf{C}_2 \right] \right)
 \end{aligned}$$

Null space through intersections

$$\begin{aligned}
 & \left[1 \mid x \ y \mid x^2 \ xy \ y^2 \mid x^3 \ x^2y \ xy^2 \ y^3 \right] \\
 \text{null}(\mathbf{M}_3) = & \text{null} \left(\left[\begin{array}{ccc|ccc|cccc} 1 & 2 & 3 & 1 & 5 & 6 & 0 & 0 & 0 & 0 \\ 2 & 1 & 7 & 8 & 1 & 9 & 0 & 0 & 0 & 0 \\ \hline 0 & 1 & 0 & 2 & 3 & 0 & 1 & 5 & 6 & 0 \\ 0 & 2 & 0 & 1 & 7 & 0 & 8 & 1 & 9 & 0 \\ 0 & 0 & 1 & 0 & 2 & 3 & 0 & 1 & 5 & 6 \\ 0 & 0 & 2 & 0 & 1 & 7 & 0 & 8 & 1 & 9 \end{array} \right] \right) \\
 = & \text{null} \left(\left[\begin{array}{ccc|ccc|cccc} 1 & 2 & 3 & 1 & 5 & 6 & 0 & 0 & 0 & 0 \\ 2 & 1 & 7 & 8 & 1 & 9 & 0 & 0 & 0 & 0 \end{array} \right] \right) \\
 \cap & \text{null} \left(\left[\begin{array}{ccc|ccc|cccc} 0 & 1 & 0 & 2 & 3 & 0 & 1 & 5 & 6 & 0 \\ 0 & 2 & 0 & 1 & 7 & 0 & 8 & 1 & 9 & 0 \end{array} \right] \right) \\
 \cap & \text{null} \left(\left[\begin{array}{ccc|ccc|cccc} 0 & 0 & 1 & 0 & 2 & 3 & 0 & 1 & 5 & 6 \\ 0 & 0 & 2 & 0 & 1 & 7 & 0 & 8 & 1 & 9 \end{array} \right] \right).
 \end{aligned}$$

Null space through intersections

[illegible]

Algorithm 2 MINI: Macaulay Iterative Null space Intersection

Require: Set of polynomials Σ of degree D in N unknowns

Ensure: Macaulay null space at degree E

```
1:  $\mathbf{K}_D \leftarrow \text{null}(\text{coeff}(\Sigma))$   
2: for  $d = D + 1$  to  $E$  do  
3:    $\mathbf{K}_d \leftarrow \text{shift}(1, \mathbf{K}_{d-1})$   
4:   for  $n = 1$  to  $N$  do  
5:      $\mathbf{K}_d \leftarrow \mathbf{K}_d \cap \text{shift}(x_n, \mathbf{K}_{d-1})$   
6:   end for  
7: end for
```

Outline

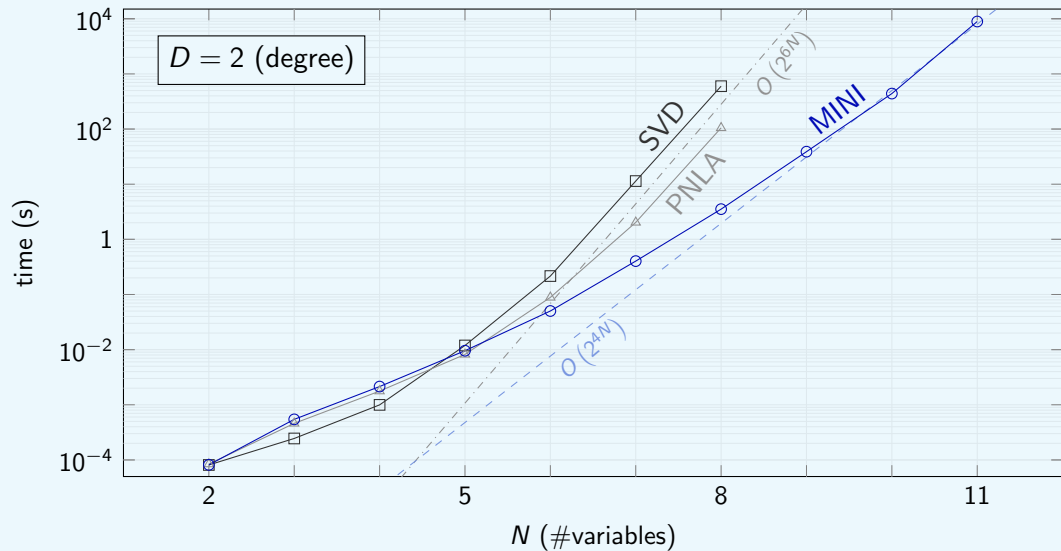
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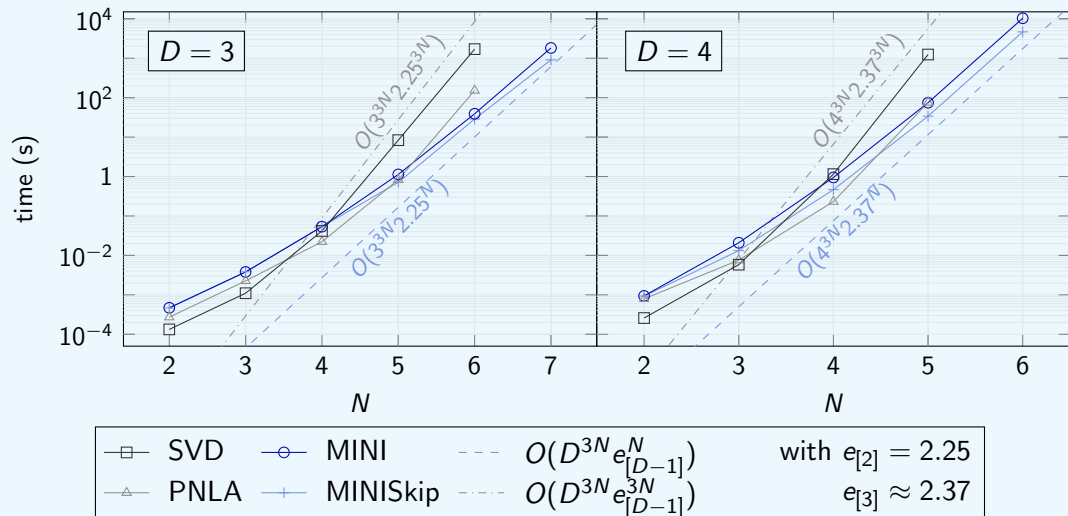
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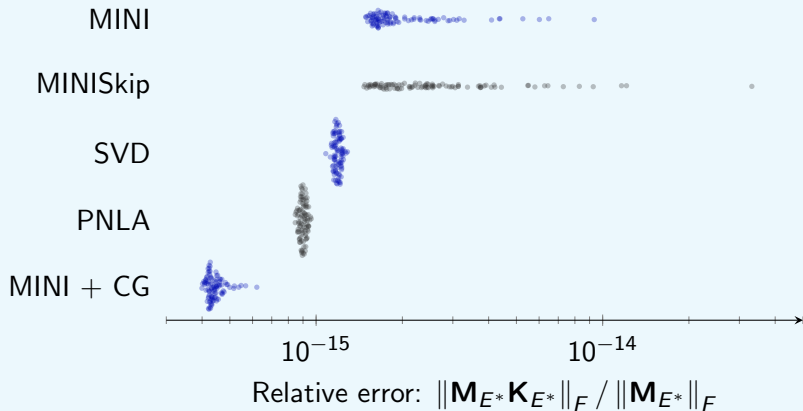
Timing results



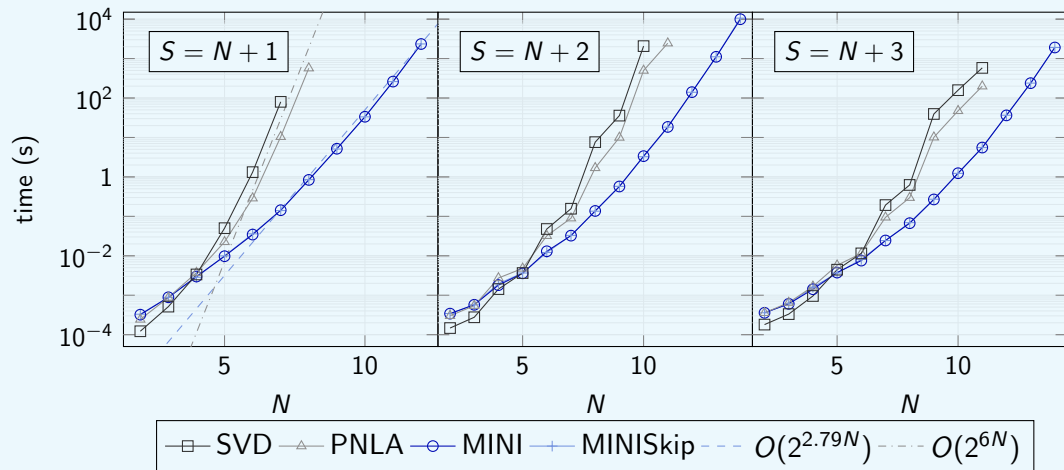
Timing results



Accuracy results



Timing results overdetermined systems



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